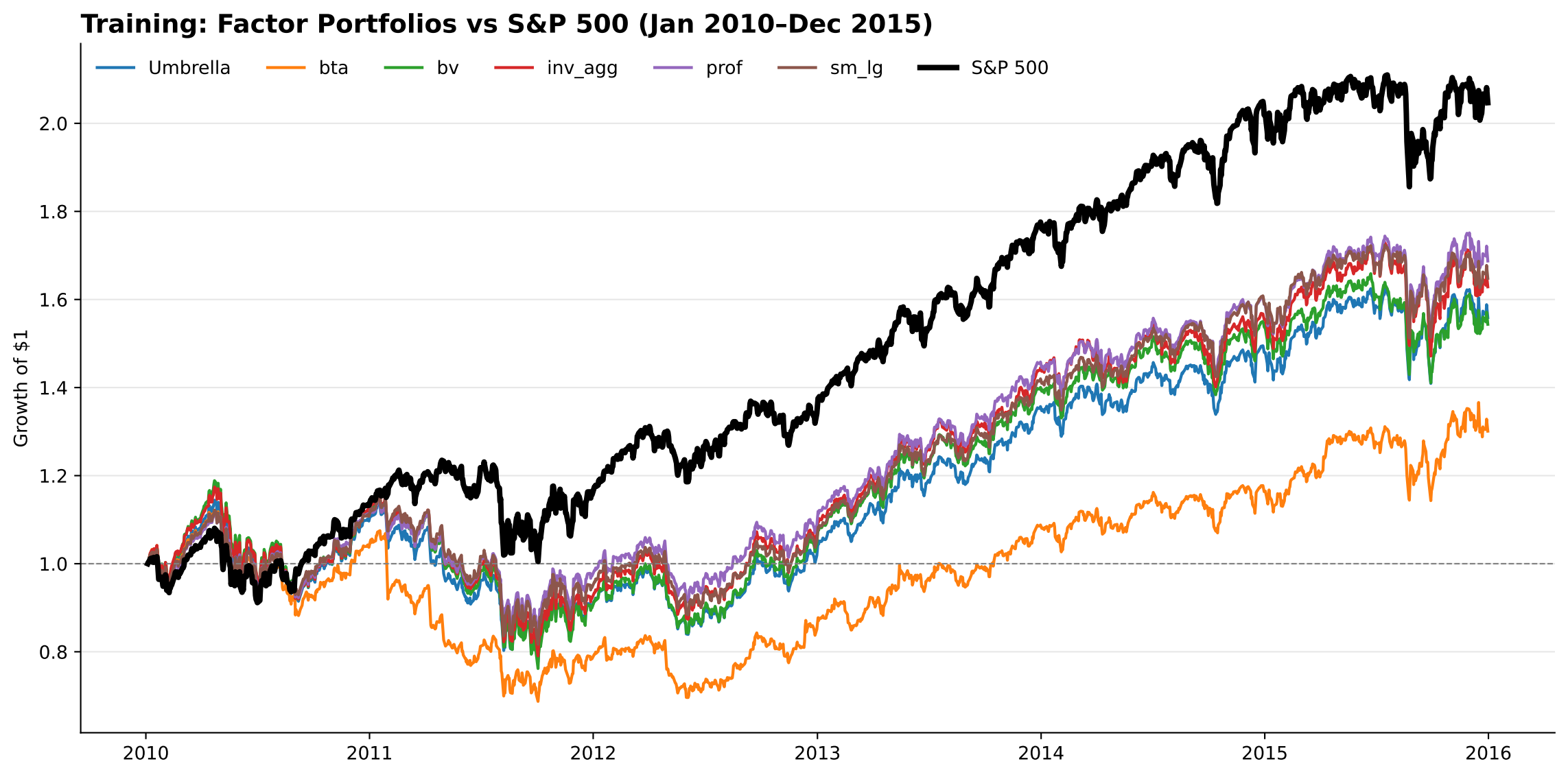
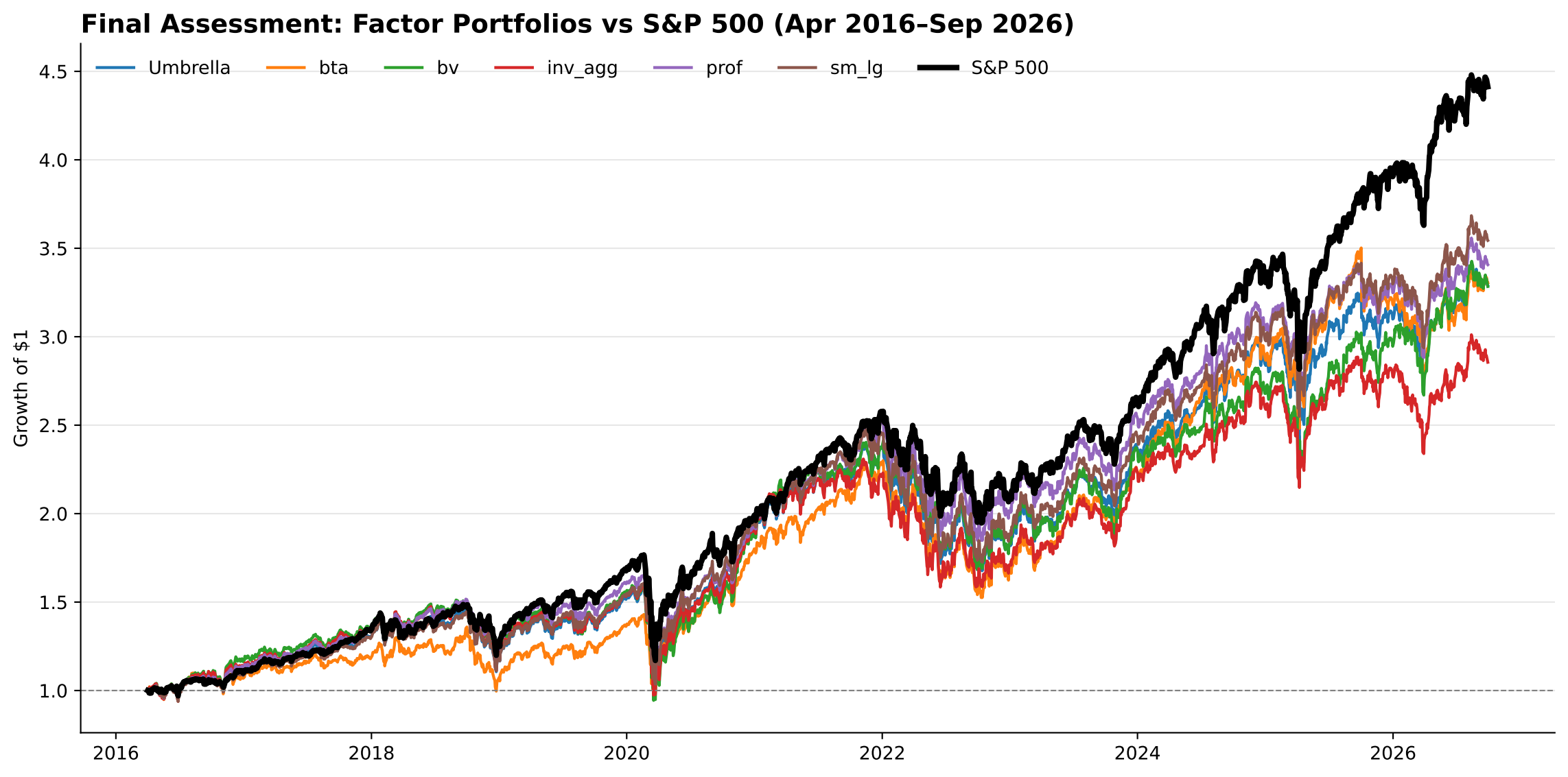

Training: Factor Portfolios vs S&P 500 (Jan 2010–Dec 2015)
Umbrella
bta
bv
inv_agg
prof
sm_lg
S&P 500
Growth of $1
2.0
1.8
1.6
1.4
1.2
1.0
0.8
2010
2011
2012
2013
2014
2015
2016

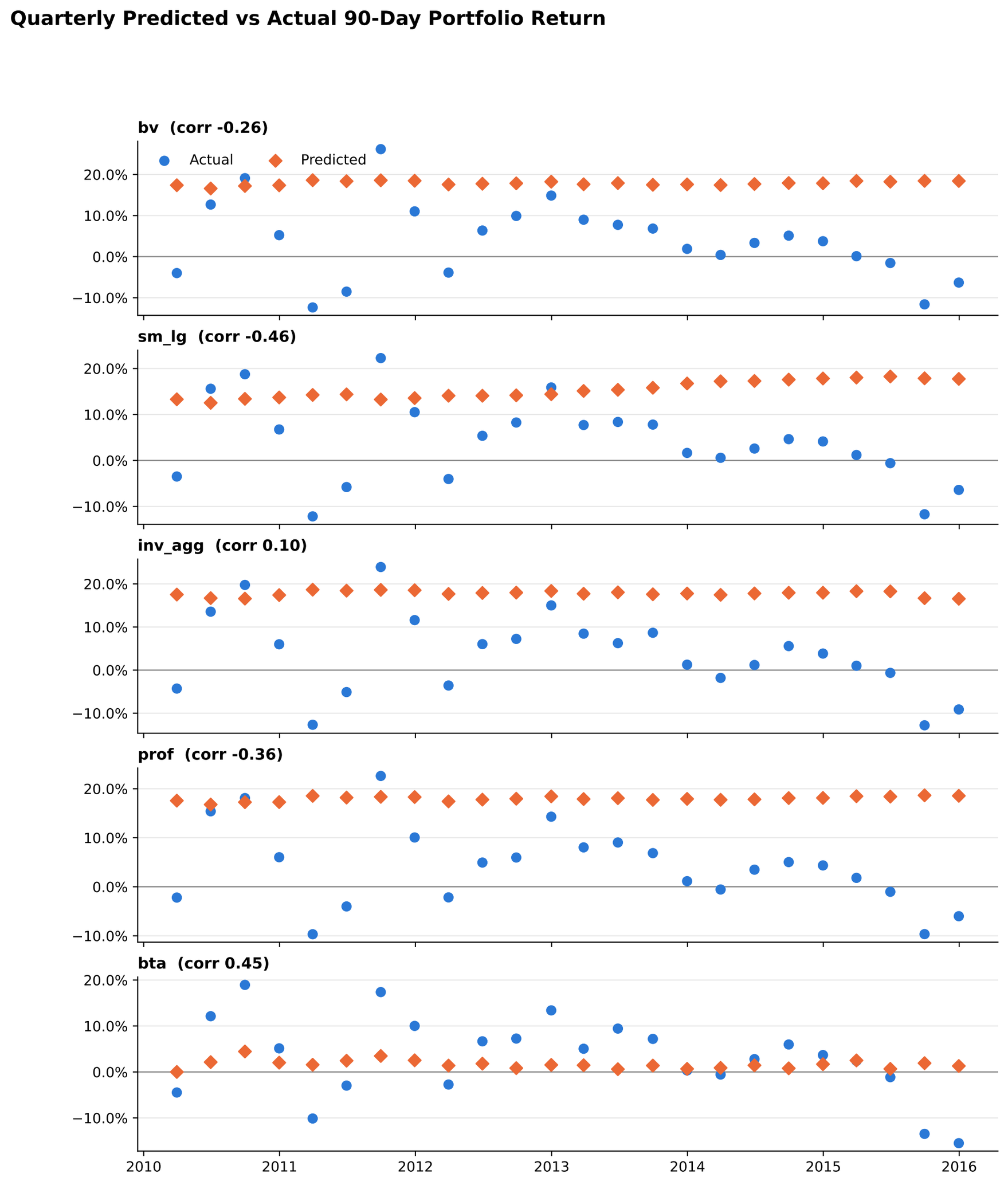
Train Period: Assessment

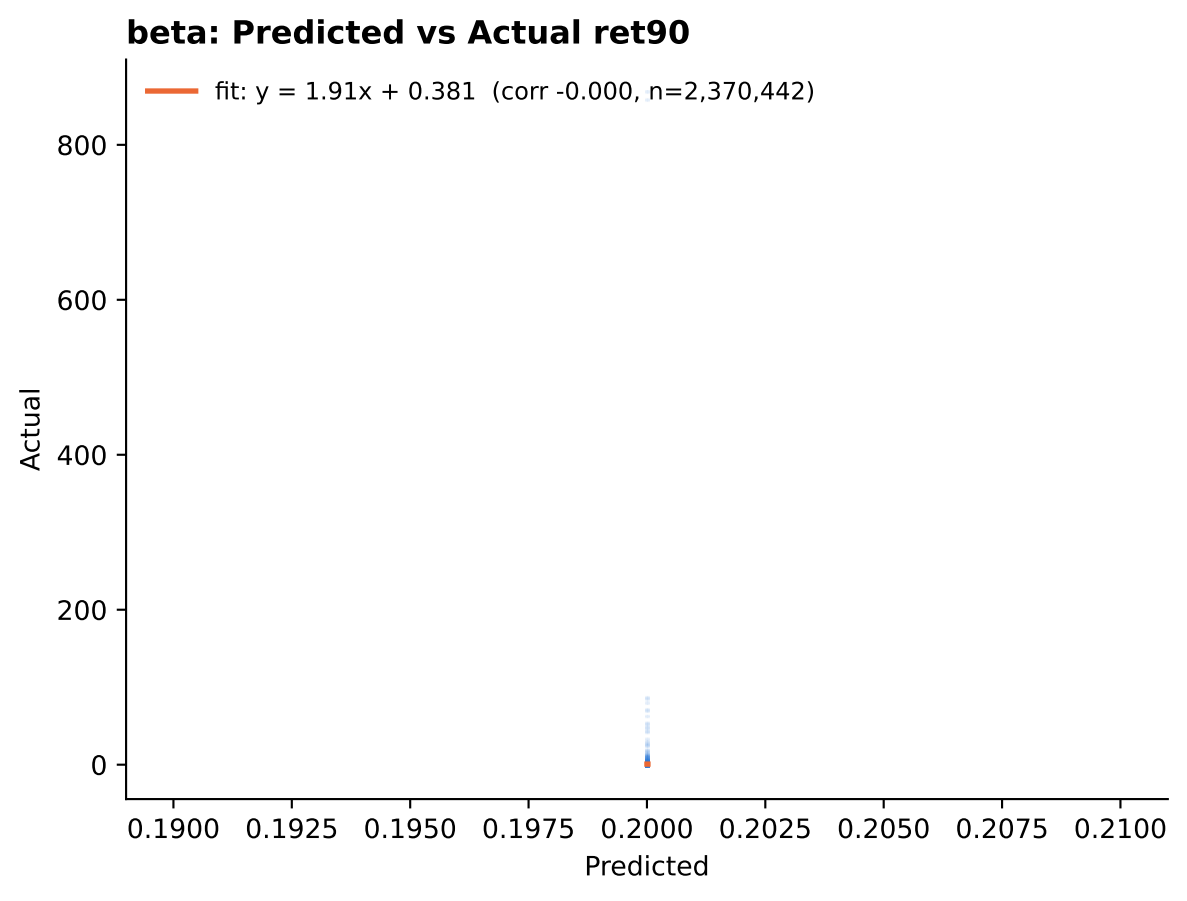
	CAGR	Vol	IR	Sharpe	SP CAGR	SP Sharpe
Umbrella	7.70%	17.90%	-0.51	0.39	12.71%	0.71
bta	4.49%	18.71%	-0.62	0.22	12.71%	0.71
bv	7.52%	18.64%	-0.50	0.38	12.71%	0.71
inv_agg	8.49%	18.43%	-0.39	0.43	12.71%	0.71
prof	9.13%	17.27%	-0.41	0.48	12.71%	0.71
sm_lg	8.70%	17.48%	-0.52	0.45	12.71%	0.71

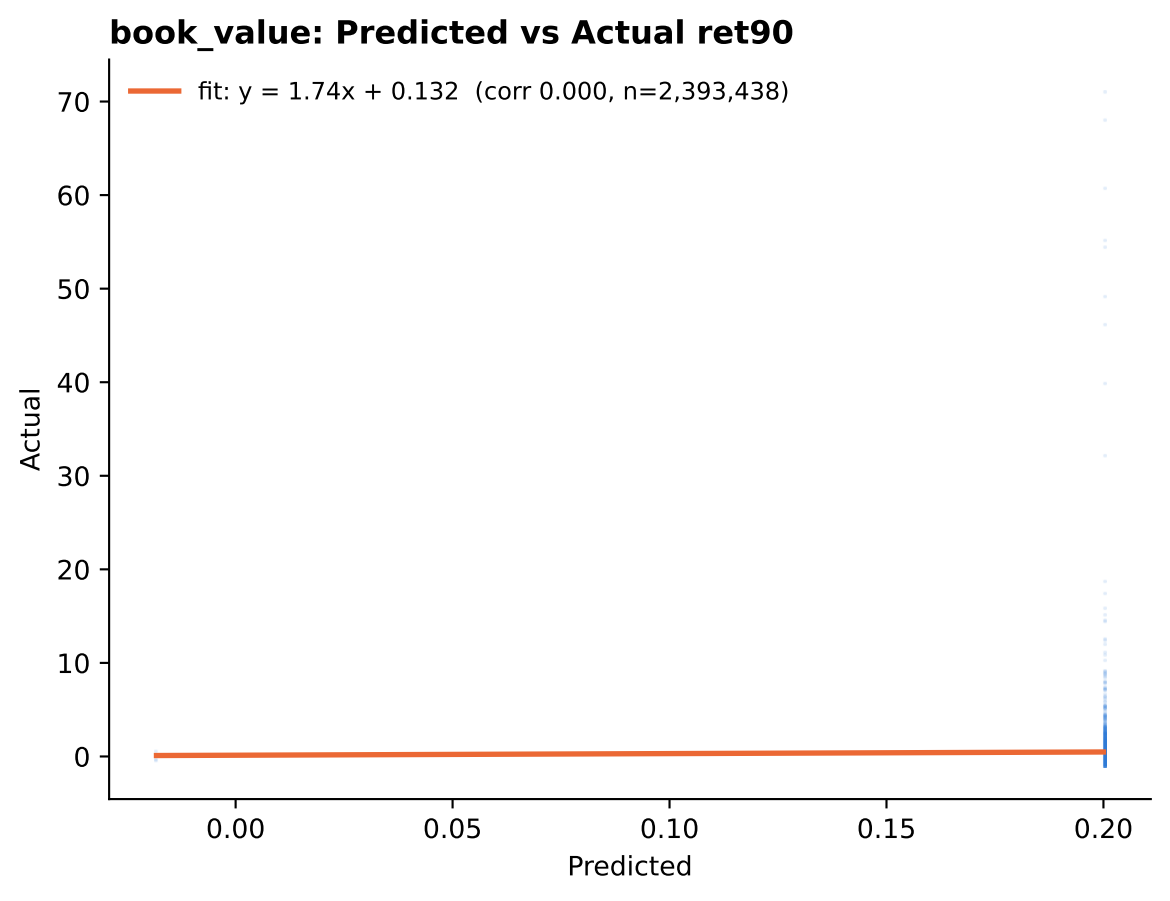

Final Assessment: Factor Portfolios vs S&P 500 (Apr 2016–Sep 2026)
Umbrella
bta
bv
inv_agg
prof
sm_lg
S&P 500
Growth of $1
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
2016
2018
2020
2022
2024
2026

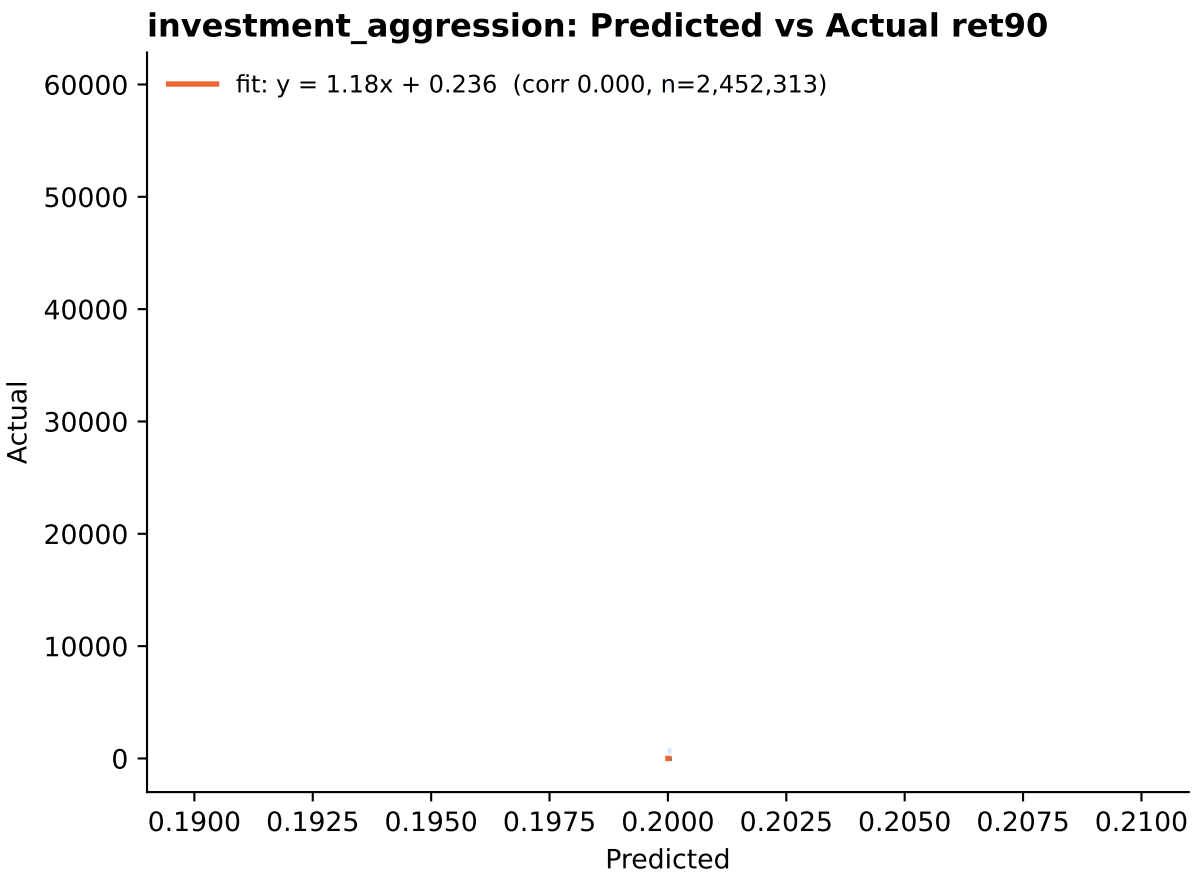
Test Period: Assessment

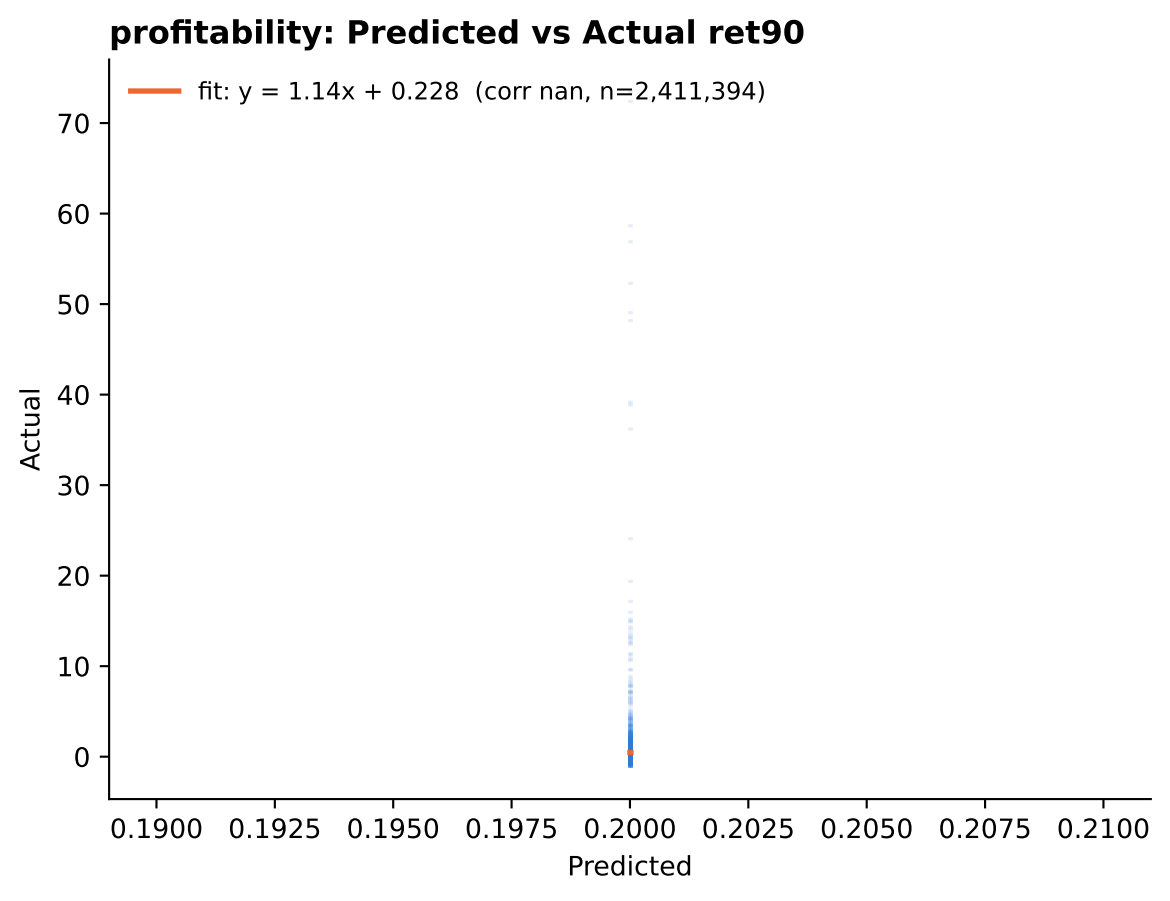
	CAGR	Vol	IR	Sharpe	SP CAGR	SP Sharpe
Umbrella	12.01%	19.91%	-0.37	0.57	15.20%	0.77
bta	12.07%	19.78%	-0.32	0.58	15.20%	0.77
bv	12.00%	21.49%	-0.24	0.54	15.20%	0.77
inv_agg	10.52%	20.83%	-0.43	0.49	15.20%	0.77
prof	12.39%	19.29%	-0.38	0.60	15.20%	0.77
sm_lg	12.81%	19.40%	-0.35	0.62	15.20%	0.77

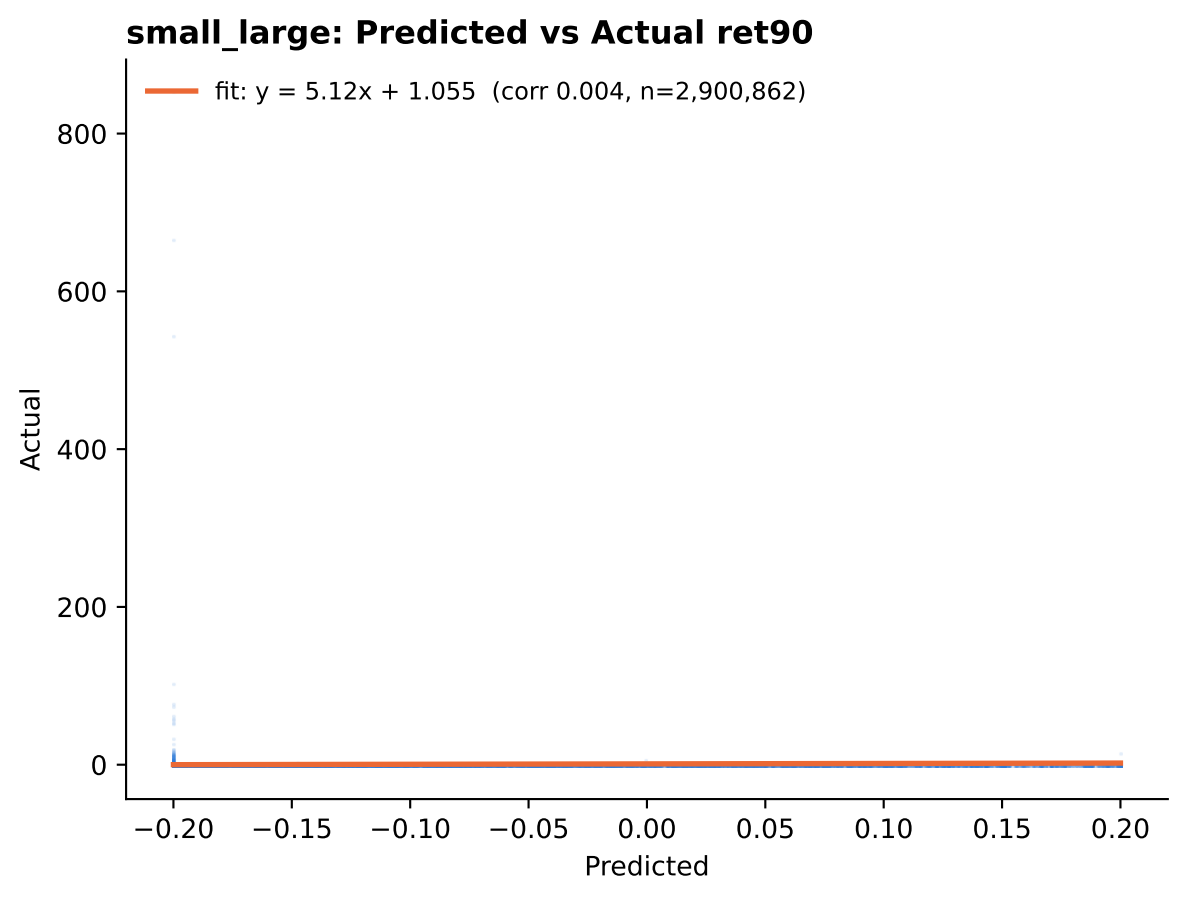

Quarterly Predicted vs Actual 90-Day Portfolio Return
bv (corr -0.26)
Actual
Predicted
sm_lg (corr -0.46)
inv_agg (corr 0.10)
prof (corr -0.36)
bta (corr 0.45)
20.0%
10.0%
0.0%
−10.0%
2010
2011
2012
2013
2014
2015
2016


beta: Predicted vs Actual ret90
fit: y = 1.91x + 0.381 (corr -0.000, n=2,370,442)
800
600
400
200
0
Actual
0.1900
0.1925
0.1950
0.1975
0.2000
0.2025
0.2050
0.2075
0.2100
Predicted


book_value: Predicted vs Actual ret90
fit: y = 1.74x + 0.132 (corr 0.000, n=2,393,438)
70
60
50
40
30
20
10
0
Actual
0.00
0.05
0.10
0.15
0.20
Predicted


investment_aggression: Predicted vs Actual ret90
fit: y = 1.18x + 0.236 (corr 0.000, n=2,452,313)
60000
50000
40000
30000
20000
10000
0
Actual
0.1900
0.1925
0.1950
0.1975
0.2000
0.2025
0.2050
0.2075
0.2100
Predicted


profitability: Predicted vs Actual ret90
fit: y = 1.14x + 0.228 (corr nan, n=2,411,394)
Actual
70
60
50
40
30
20
10
0
0.1900
0.1925
0.1950
0.1975
0.2000
0.2025
0.2050
0.2075
0.2100
Predicted


small_large: Predicted vs Actual ret90
fit: y = 5.12x + 1.055 (corr 0.004, n=2,900,862)
800
600
400
200
0
Actual
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
Predicted